

Command sequence in "main_interfaz.f90"

This document describes the sequence of operations performed by "main_interfaz" (source file "main_interfaz.f90"). When the executable is run, the user is guided through the steps below via a series of prompts in the terminal. Inputs are read from standard input; all error/EOF conditions are handled gracefully and any non-positive time step is rejected. The sequence of operations is:

1. Pre-process: query IEEE floating-point support and, if available, disable underflow/inexact/overflow/divide-by-zero/invalid exception flags so that benign exceptions do not abort the run.
2. Print introductory message: **"Esta es la interfaz para resolver la parte reactiva de una iteración de mezcla reactiva en 1D usando Euler explícito."**
3. Create chemistry class object: **my_chem**
4. Read directory of chemical databases (**dir_DB**), including a trailing backslash or forward slash depending on the OS.
5. Read directory of the problem to be solved (**dir_pb**), with the corresponding trailing slash.
6. Read the common root of the input and output file names (**root_files**).
7. Read chemistry: **my_chem** calls **read_chemistry_interface**, passing the trimmed root, problem directory and database directory.
8. Read the name of the file where **my_chem** will write the aqueous component concentrations of the initial and external water types (**file_u_wat_types**).
9. Write those concentrations: **my_chem** calls **write_conc_comp_wat_types**.
10. Read the name of the file where **my_chem** will write the concentrations after each reactive mixing iteration (**file_u_new**).
11. Read the name of the file containing the aqueous component concentrations after solving an iteration of conservative transport (**file_u_tilde**). The file must live in the problem directory. Rows correspond to components and columns to targets.
12. Read the initial time step (**Delta_t**). Execution is aborted with an error if the value is not strictly positive.
13. Read whether the time step is constant (**flag_Delta_t** = 1) or variable (**flag_Delta_t** = 0).
14. Get the number of aqueous components in the domain: **my_chem** calls **get_num_aq_comps_tar_wat**. The value is stored in **num_aq_comps** and printed to the terminal.
15. Select the target of the procedure pointer **p_interfaz** depending on the chemistry: **interfaz_esp_arch** if there are no equilibrium reactions, **interfaz_comps_arch** otherwise. The chosen target is the reactive mixing interface used in the loop.
16. Print a reminder that the file with the conservative transport concentrations must be updated by the user before each iteration.
17. Reactive mixing loop:
 - Compute one reactive mixing iteration: call **p_interfaz** with **my_chem**, **dir_pb**, **num_aq_comps**, **file_u_tilde**, **Delta_t** and **file_u_new**.
 - Ask the user whether to continue iterating (**flag** = 1) or stop (**flag** = 0). Any other value aborts execution with the error **"Opción no válida. Tiene que ser 1 o 0."**
 - If the time step is variable (**flag_Delta_t** = 0) and the user chose to continue, read the new value of **Delta_t**; the new value must also be strictly positive.
18. Post-process: if IEEE support is available, clear the floating-point exception flags before exiting.
19. Print termination message **"El programa ha terminado."** and end the program.